

Beyond Immediate Evaluation: A Longitudinal Study of Learning Effects in an Educational Game

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Abstract

Educational games are commonly evaluated using immediate post-use measures, which may underestimate learning processes that unfold over time. We present an exploratory longitudinal study of a narrative-driven serious game set in Ancient Greece, designed to support historical learning through immersive interaction. Twenty undergraduate students played the game either with a virtual companion (WC) or without a companion (WOC). Knowledge was assessed before gameplay, immediately after, and again two months later. Immediate post-test results showed mixed and sometimes limited learning gains. However, delayed results revealed substantial improvement for most participants, including some who showed little or no immediate progress. These findings suggest that narrative and interactive experiences can foster implicit learning and knowledge consolidation that is not captured by short-term evaluation. The study highlights the importance of longitudinal methods in HCI research and offers design insights into the trade-offs between guided and autonomous gameplay in educational systems.

CCS Concepts

• **Human-centered computing**; • **Human-computer interaction**; • **Applied computing**; • **Education**; • **Computer games**;

Keywords

Game-based learning, longitudinal evaluation, educational games, narrative interaction, virtual companions, HCI

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1 Introduction

Educational technologies increasingly use serious games to enhance engagement, motivation, and learning, especially in STEM and history education [3, 6, 13]. In HCI, game-based learning (GBL) is not just content delivery, but a framework where learning emerges through interaction, feedback, and exploration [17, 22]. Prior work shows that well-designed games can support conceptual understanding, problem-solving, and transferable skills [16], yet most evaluations rely on immediate post-use measures, leaving long-term outcomes underexplored [4, 12, 16].

Understanding long-term knowledge retention is critical when learning involves narratives and memory consolidation. HCI research suggests that short-term evaluations often miss deeper cognitive effects [7, 11, 20]. Long-Term Working Memory (LTWM) theory indicates that information can be encoded and maintained over time [7–9], motivating longitudinal assessment in GBL.

This study examines a narrative-driven educational game set in Ancient Greece, comparing With Companion (WC) and Without Companion (WOC) conditions. Knowledge was measured immediately and two months later to assess persistence and the role of companion support. The contributions are: (i) insights into narrative-driven game design and longitudinal evaluation in HCI, and (ii) evidence on the impact of virtual companions on engagement and retention.

2 Related work

2.1 Game-Based Learning in HCI

Game-based learning (GBL) has gained attention in HCI and computing education for supporting engagement and motivation through interactive experiences. These games are widely used in STEM, particularly in programming, where abstract concepts challenge learners [3, 13]. From an HCI perspective, learning emerges in these systems through action, feedback, and situated experience.

GBL effectiveness depends on balancing educational goals with motivational affordances like challenge, competence, and feedback



Figure 1: Reconstructed environment of Ancient Athens and in-game characters used in the educational game.

[6]. While serious games in programming show high engagement and perceived benefits, empirical outcomes can be inconclusive [17, 22].

Across domains, GBL fosters high motivation even when immediate knowledge gains are modest [10, 16]. However, reliance on short-term evaluations and small samples limits insights into durability [4, 16]. Prior work suggests GBL supports processes that unfold over time, such as conceptual understanding and transferable skills [4, 12].

2.2 Working Memory, Interruptions, and Cognitive Mechanisms in HCI

HCI research has historically followed two paths: improving usability through design and user testing [11, 20] and explaining the cognitive mechanisms of user psychology [21]. Within the latter, Long-Term Working Memory (LTWM) serves as a key framework for understanding how users retain and retrieve task-relevant information during extended interactions or interruptions [7].

Working memory consists of Short-Term Working Memory (STWM) and LTWM [7–9]. While STWM is limited in capacity and supports attentional control [1, 5], LTWM allows STWM information to be encoded into long-term memory via structured retrieval cues, facilitating rapid access based on expertise [7, 9].

Interruptions in interactive environments often disrupt STWM but leave LTWM more resilient [7, 19], though semantically similar interruptions can cause interference [2, 23]. These findings highlight the limitations of short-term usability evaluations, which may overlook consolidation effects [7, 11, 20]. Incorporating LTWM into HCI evaluation requires accounting for interruptions, long-term learning outcomes, and the differences between novice and expert users to better capture durable retention.

3 Game overview

3.1 Game Background

The game combines RPG and Adventure genres for their broad gender appeal [14] and narrative-driven educational utility. Its realistic 3D environment facilitates spatial engagement with Ancient Athens to aid retention. To ensure immersion, landmarks and characters, including clothing and aesthetics, were rigorously reconstructed using historical references. The following figure, Fig 1, represents an example of how the design of buildings and places were done and shows the characters designed.

3.2 Game Narrative

Narrative facilitates immersive learning [18]. Set during a Persian attack, players search for missing family across five historically integrated missions. The game compares WC and WOC conditions. In WC, a wounded survivor acts as an ally for cooperative challenges. In WOC, the player acts independently; game mechanics, including puzzles and combat difficulty, adapt to ensure a balanced but autonomous experience.

4 Methodology

4.1 Study Design

We conducted a controlled between-subjects study to evaluate learning outcomes in an educational serious game developed as part of a master's thesis. The study compared two experimental conditions: WC, in which players interacted with an in-game companion providing narrative and instructional support, and WOC, in which the same game content was presented without the companion. The primary goal was to examine whether the presence of a companion influenced short-term and long-term knowledge retention.

4.2 Participants

The study involved 20 undergraduate students from technical degree programs at local university. While all possessed technical backgrounds, none had formal instruction in the game's specific historical content. Participants were randomly assigned to two groups: 10 in the WC condition and 10 in the WOC condition.

Although all 20 participants completed the initial experiment, the sample decreased to 13 for the two-month follow-up questionnaire. Participation was voluntary, and anonymity was maintained by having participants create unique IDs for data tracking.

4.3 Procedure

Each participant played the game individually under their assigned condition (WC or WOC). Before gameplay, a pre-test (PRE) established baseline knowledge. Immediately afterward, participants completed a post-test (POST) with the same items to measure immediate gains.

To evaluate long-term retention, the questionnaire was administered again two months later. This delayed assessment was conducted on campus under supervision to prevent the use of external resources. Participants were not informed of the follow-up in advance, minimizing potential rehearsal or revision effects.

Table 1: Scores trajectories across PRE, POST, and Delayed POST measurements

	PRE	POS	Difference	After 2 months	Difference
WC	30	50	20	75	45
WOC	18,75	31,25	12,5	57,14	38,39

4.4 Measures

Learning outcomes were measured using a knowledge-based questionnaire of factual and conceptual items, with scores expressed as percentages. Data were collected at three points: PRE (baseline), POST (immediate), and Delayed POST (two months later). Learning gains were determined by comparing PRE–POST scores, while retention was assessed via POST–Delayed scores. The Delayed POST also included an adapted Epistemic Curiosity Scale [15] regarding Ancient Greece. This scale, measuring the desire for knowledge, retained its Interest-type and Deprivation-type subscales. Items were translated and reworded to fit the serious game context.

4.5 Data Analysis

Descriptive analyses examined score changes across three time points in the WC and WOC conditions. Individual trajectories were calculated to observe learning gains, stability, or declines. Because long-term retention is often underrepresented in HCI game evaluations, delayed performance was prioritized despite the limited sample size. Using identical questionnaires ensured consistent comparisons, while administering tests in supervised academic settings reduced threats to internal validity, such as cheating or uncontrolled rehearsal.

5 Results

5.1 Knowledge Outcomes

We examined knowledge scores across three time points: PRE (baseline), POST (immediately after gameplay), and DELAY (two months later). Scores were calculated as percentage of correct responses (0–100%). Visual analysis (see Table 1) illustrates score trajectories across PRE, POST, and Delayed POST measurements, highlighting individual differences and the effect of companion presence on learning patterns.

5.1.1 Immediate Learning Gains (PRE vs POST). Across all participants ($N = 20$), a Wilcoxon signed-rank test revealed a significant increase from PRE to POST, $W = 13.50$, $p = .038$, with a medium-to-large effect size ($r = .46$). These results indicate that gameplay produced statistically significant immediate learning gains. Descriptively, most participants improved their scores following the intervention, although individual trajectories varied. Improvements were observed in both WC and WOC conditions.

5.1.2 Delayed Retention (PRE vs DELAY). Thirteen participants completed the two-month follow-up assessment. A Wilcoxon signed-rank test comparing PRE and DELAY scores did not reach statistical significance, $W = 0.00$, $p = .102$. However, the associated effect size was large ($r \approx .61$), suggesting meaningful long-term trends that may not have reached statistical significance due to the reduced follow-up sample size. Descriptively, several participants

demonstrated maintained or improved performance at DELAY relative to POST, indicating that learning effects persisted over time for part of the sample.

These findings align with cognitive frameworks of STWM and LTWM, where the companion helps manage STWM load and facilitates encoding into LTWM, while independent play imposes greater cognitive demands, which can hinder immediate performance but enhance durable knowledge retention. Overall, the results highlight the value of interactive, narrative-driven gameplay for supporting both immediate and delayed learning, emphasizing the need for longitudinal assessment in educational game evaluation [4, 7].

5.2 Epistemic Curiosity Scale

At the two-month follow-up ($N = 13$), participants completed an adapted Epistemic Curiosity Scale [15]. Interest-type curiosity showed a mean of $M = 3.49$ ($SD = 1.15$), whereas Deprivation-type curiosity was slightly lower at $M = 3.03$ ($SD = 1.14$) on a 5-point scale. Between-condition differences were minimal WC: Interest $M = 3.53$; Deprivation $M = 2.97$; WOC: Interest $M = 3.46$; Deprivation $M = 3.09$. Table 2 displays the results. These findings suggest that while immediate learning effects varied, both conditions resulted in sustained baseline interest in the historical topic two months after gameplay.

6 Discussion

This study investigated whether a narrative-driven educational game supports immediate and delayed learning, and whether a virtual companion influences these outcomes. Results showed statistically significant knowledge gains from PRE to POST, with a medium-to-large effect size ($r = .46$), indicating that even a single gameplay session produced measurable learning improvements.

The delayed assessment showed strong descriptive improvements relative to PRE scores, although these differences were not statistically significant. The large PRE–DELAY effect size ($r \approx .61$) suggests the change may still be meaningful despite limited statistical power due to the smaller follow-up sample. This pattern indicates that some participants maintained or consolidated knowledge over time. While traditional forgetting curves predict decline, narrative structure may provide retrieval cues that support longer-term recall, though these cognitive mechanisms were not directly measured.

Two months after gameplay, participants reported moderate levels of Interest-type epistemic curiosity ($M = 3.49$) and slightly lower Deprivation-type curiosity ($M = 3.03$), with minimal differences between conditions. This suggests the game may have supported a sustained baseline interest in Ancient Greece even after immediate knowledge gaps were resolved. Although curiosity cannot be attributed solely to gameplay, the follow-up responses indicate the experience did not lead to disengagement from the topic.

Table 2: Results from the Interest-type and Deprivation-type. It includes the mean and standard deviation (SD) of WC and WOC separated and when combined.

Subscale	Group	N	Mean	SD
Interest-type (I-type)	WC	6	3.53	0.88
	WOC	7	3.46	1.40
	Total	13	3.49	1.15
Deprivation-type (D-type)	WC	6	2.97	0.73
	WOC	7	3.09	1.46
	Total	13	3.03	1.14

7 Conclusion and Future Work

This study examined learning outcomes in a narrative-driven educational game set in Ancient Greece, comparing gameplay with and without a virtual companion across immediate and delayed assessments. Results showed statistically significant immediate knowledge gains after gameplay, with a medium-to-large effect size ($r = .46$), indicating that even a single session supported measurable learning.

Although delayed improvements were not statistically significant, the large effect size ($r \approx .61$) suggests meaningful long-term learning trends that should be explored with larger samples. The longitudinal assessment also revealed retention patterns that would not have been visible through immediate post-testing alone.

Follow-up curiosity measures indicated sustained interest in the historical topic two months after gameplay, suggesting that narrative-driven educational games may support both knowledge acquisition and longer-term engagement. Overall, the study highlights the importance of longitudinal evaluation in HCI research on educational games and the value of reporting both statistical significance and effect sizes in small-sample studies. Future work should expand sample sizes, use parallel test forms to reduce repeated-measure effects, and include process-based measures such as cognitive load or in-game behavior analytics.

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